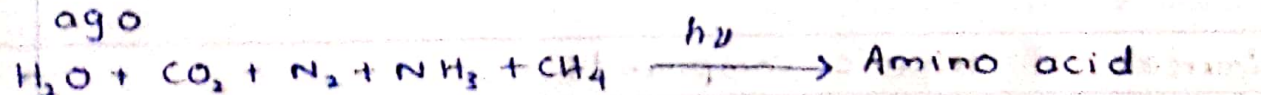


Introduction

Organic chemistry is in chemistry of compound which contain the element carbon (C). These are originated by living organisms. Most of these compounds contain (98%) H_2 and many also contain O_2 , N_2 or other elements. Some carbon-containing compounds are not classified as organic, for example, carbonates and cyanides and simple oxides, such as CO and CO_2 .

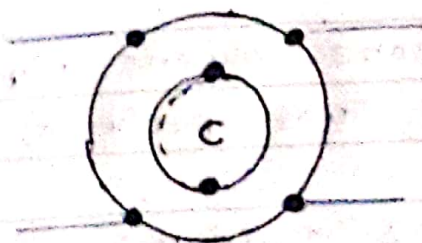
First Organic Molecule.

- The first organic molecule formed 4 billion years ago



Features

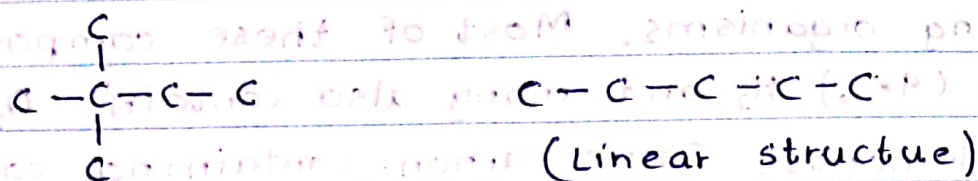
'C' have 4 electrons in the outer ring, each of there can be shared with electrons from other element that are able to complete there electronic shell, by sharing electrons to form covalent bond.



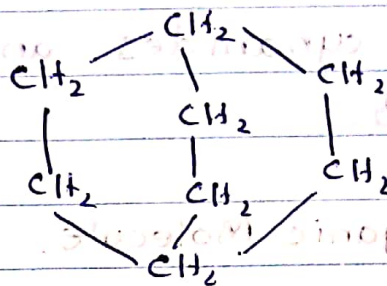
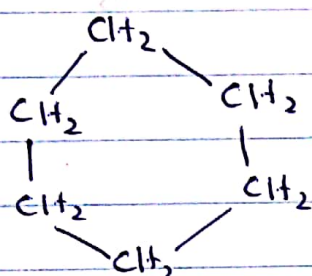
Most Significantly

- (i) 'C' can form covalent bonds with other molecules such as H_2 , N_2 , O_2

(2) 'C' can form covalent bonds with another 'C' atom. This help to form various type of biological compounds.



(branch)



Sources of organic compounds

- (1) Plants - Sugar, starch, cellulose, citric acid, oxalic acid, pterodactyl, Fats & oils, vitamins
- (2) Animals - Fats & oils, Proteins, vitamins, hormones, Urea
- (3) Wood distillation - acetic acid, methonal, Acetone
- (4) Coal distillation - Benzene, Toluene, naphthalene
Cresol, Dyes, perfume, ~~pyrid~~, Drugs
- (5) Natural gas & petrolene - Alkanes, Alkynes
distillation
- (6) Fermentation process - Ethonal, Acetic Acid, Amine

Applications of Organic compounds

- (1) Food - cereals, vegetables, fats, oil, sugar

Lipids

[In room temperature oil is liquid and fats are solid]

- (2) Clothing - Cotton, wool, silk, nylon

- (3) Medicine - Aspirin, Quinine, Sulphur drugs, peniciline

Nomenclature

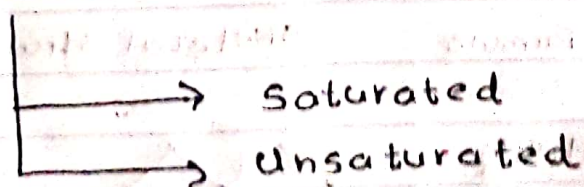
The nomenclature is classified accordance with two main classes of compounds as aliphatic and aromatic. These compounds are further classified based on functional group of the compound.



ഫിങ്ക്ഷണൽ ഗ്രൂപ്പ്

Hydrocarbons

The simplest organic compounds contain only the elements 'C' and 'H' and are called hydrocarbons. We can divide it in two categories.



Saturated Hydrocarbons -

Hydrocarbons in which fully satisfied and therefore cannot take up more hydrogen

These saturated hydrocarbon contain 'C' atoms linked by single covalent bond or bonds

Ex:- Alkanes (CH_4 , C_2H_6 , C_5H_{12} ...)

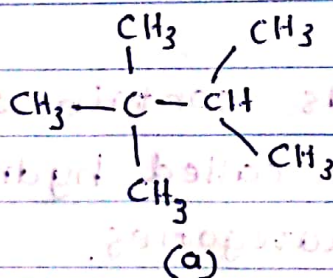
- Unsaturated hydrocarbons -

Hydrocarbons contain less hydrogen per carbon atom and the valence of 'C' atom in their molecule are not fully satisfied and therefore can take up more 'H' molecule. The unsaturated hydrocarbons are characterized by the presence of double bonds and triple bonds.

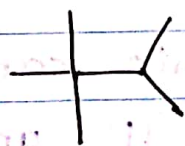
Ex:- Alkene, Alkyne

~~* A common method used by~~

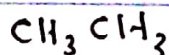
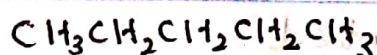
Skeletal structure :- In this type of structure, 'C' atoms are not symbolized with a 'C', but represented by each end of a line or bend in a line.



condensed formula

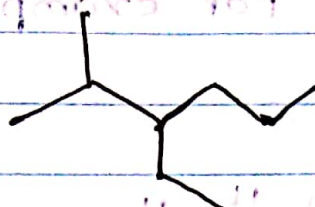
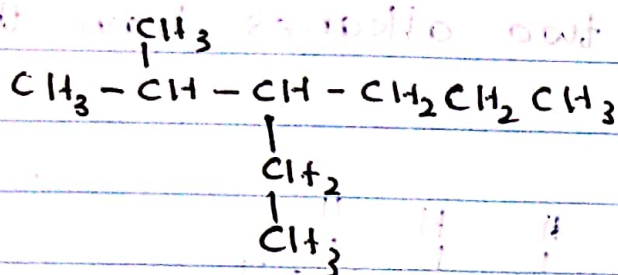
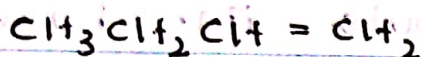
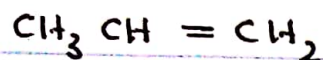
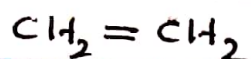


skeletal structure



No: _____

Date: ____/____/____

condensed formulaskeletal structure

No. of 'C' atoms

Prefix

Name

1

meth

methane

2

eth

ethane

3

prop

propane

4

but

butane

5

pent

pentane

6

hex

hexane

7

hept

heptane

8

oct

octane

9

non

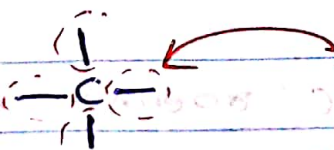
nonane

10

dec

decane

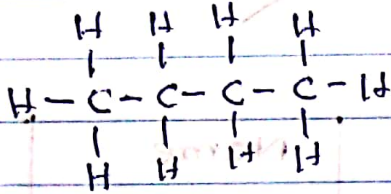
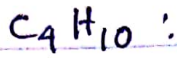
NOTE :



This means the CH_3 has
connected to the 'C' atom

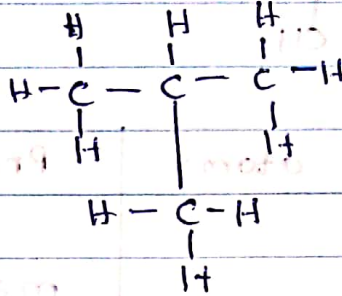
Isomers - (ଅତ୍ୟବସ୍ଥା)

Hydrocarbons with same formula, can have different structures. For example, two alkanes have the formula

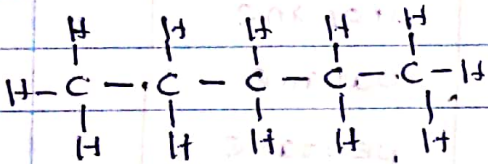


n-Butane

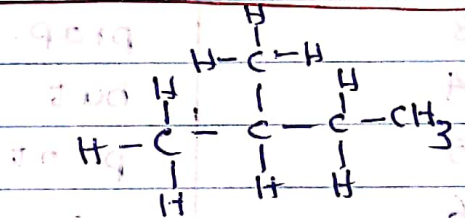
(normal-butane)



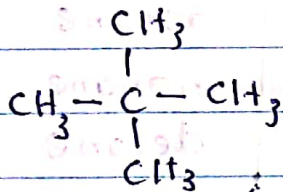
(Iso-butane)



(n-pentane).



(Iso-pentane)



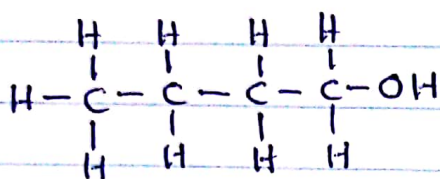
(Neo-pentane)

NOTE:-

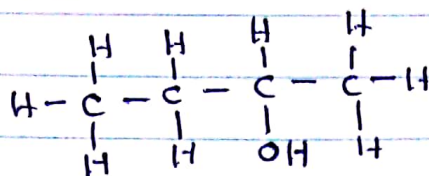
(C) රජයේ 'C' පරමාණු ගණන අනුව
අදින්නා ප්‍රචලන් ආමාණයවික ගාන

(C) පරමාණු	Isomers
①	1
②	1
③	2
④	4
⑤	8

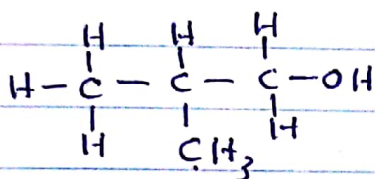
Example:- If we consider $C_4H_{10}O$, we can draw only 4 Isomers to it. Since there are 4 carbon atoms on it.



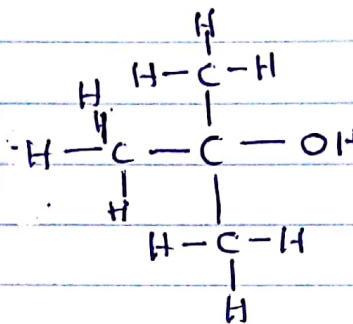
(n-butanol)



(sec-butanol)



(Iso-butanol)



(tert-butanol)

* We use the term-'normal' or the prefix n , to refer to a chain of carbon atoms without branching.

Identifying isomers from Lewis structures is not as easy as it looks. Lewis structures that look different may actually represent the same isomers. For example, the three structures in Figure 20.4 all represent the same molecule, n -butane, and hence are not different isomers. They are identical because each contains an unbranched chain of four carbon atoms.

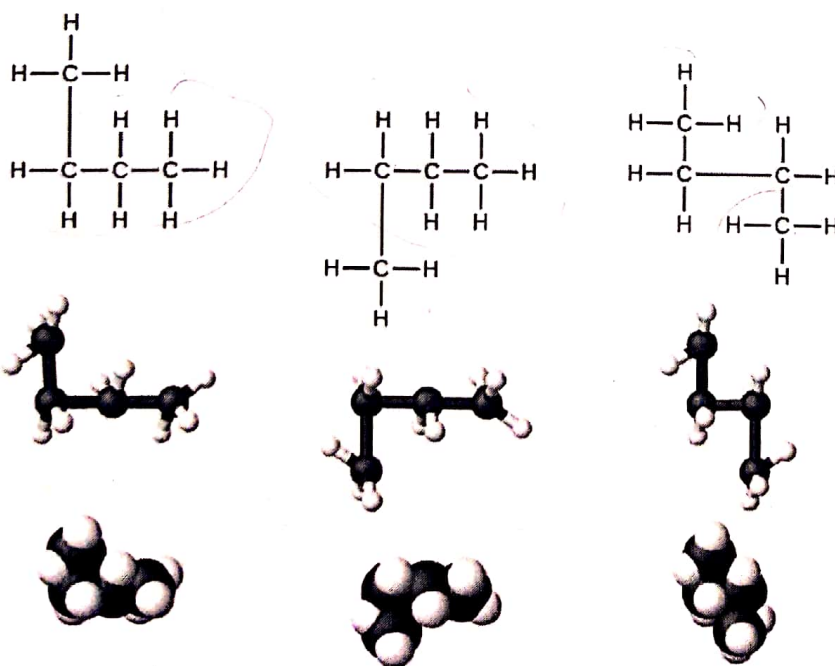


Figure 20.4 These three representations of the structure of n -butane are not isomers because they all contain the same arrangement of atoms and bonds.